Data-Poor Stock Analysis Protocol (modified CMSY)

The methods described here are based on those presented by **Giron-Nava et al. (2018)** and **Froese et al. (2017)**, using FAO-Mexico data and IUU correction factors supplied by **Cisneros-Montemayor (2013)**. The primary intent of this protocol is to help generate fisheries stock assessments for the data-poor fisheries in the Gulf of California region.

In short, the methods described here rely on a Monte Carlo estimate of fishery reference points, followed by a "*Bayesian state-space implementation of the Schaffer production model (BSM), fitted to catch and biomass or catch per unit effort (CPUE) data*" (Froese et al., 2017). To accomplish this, the primary code file takes advantage of JAGS for R (see what is JAGS?).

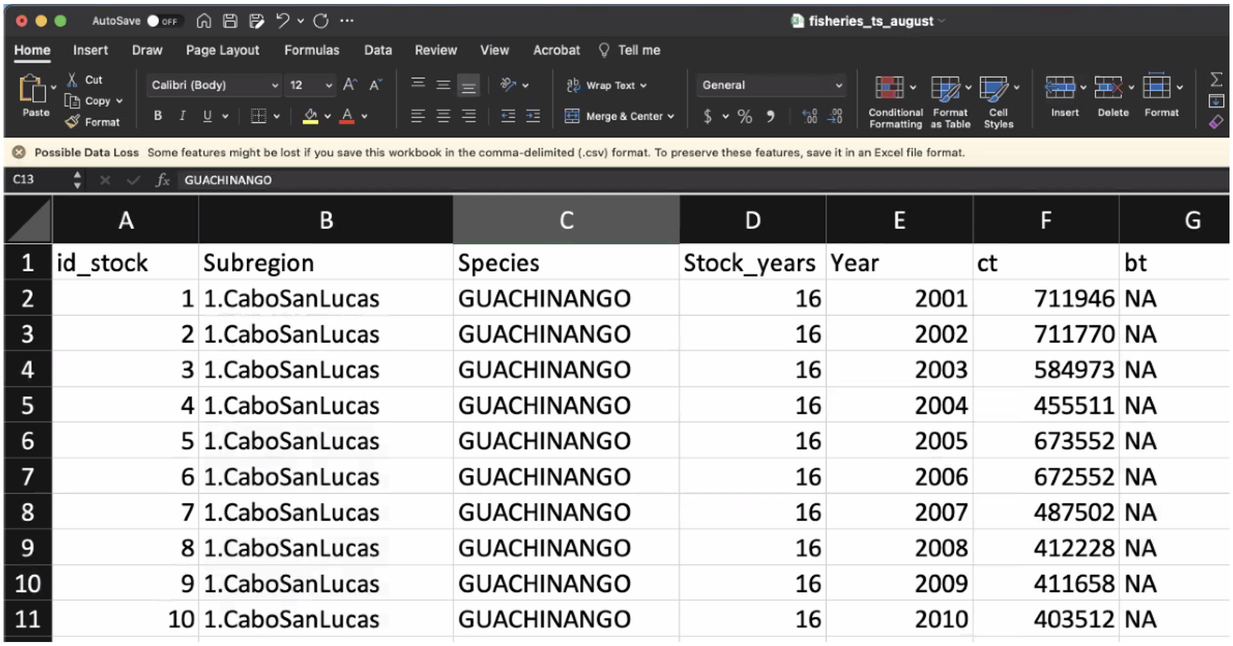
Variables:

r	Intrinsic growth of [fish] population
K	Carrying capacity
B_o	…, 2*B_{msy}
B_{msy}	Biomass at MSY (tonnes or kg)
B	Biomass (tonnes or kg)

To Begin:

You will need a couple of different files, including -- most notably -- a file with stock timeseries (**File 1**) and a "metadata" file (**File 2**), which includes information about those stocks.

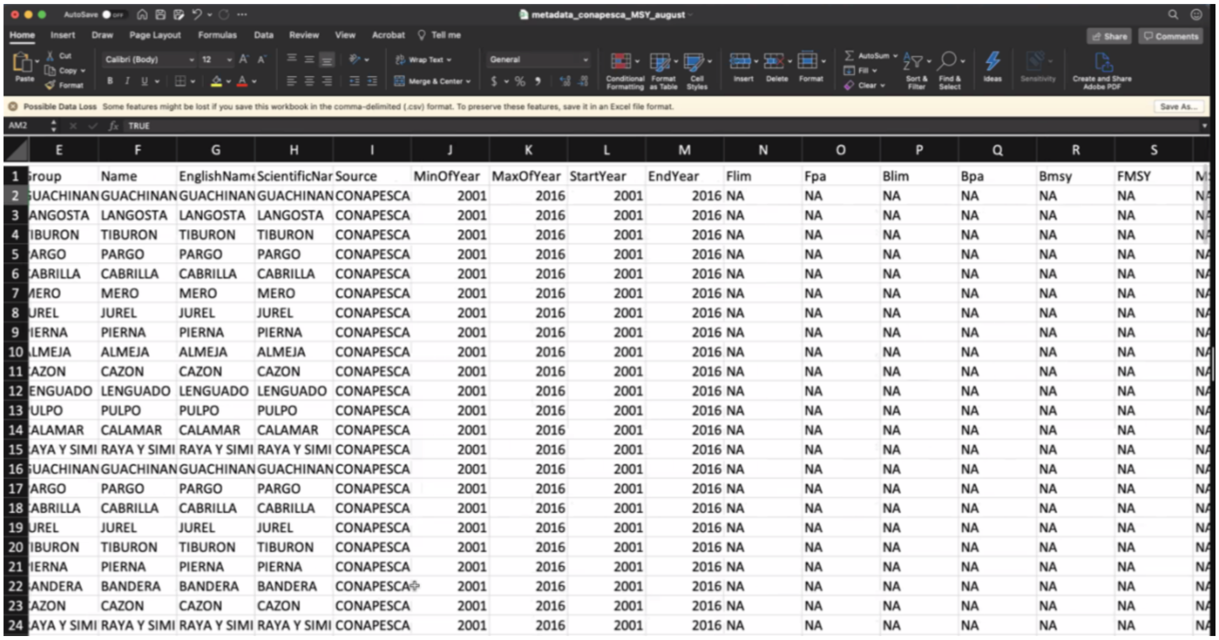
- **File 1** is a timeseries file. It will appear rather simple but it need be formatted correctly.



	A	B	C	D	E	F	G
1	id_stock	Subregion	Species	Stock_years	Year	ct	bt
2	1	1.CaboSanLucas	GUACHINANGO	16	2001	711946	NA
3	2	1.CaboSanLucas	GUACHINANGO	16	2002	711770	NA
4	3	1.CaboSanLucas	GUACHINANGO	16	2003	584973	NA
5	4	1.CaboSanLucas	GUACHINANGO	16	2004	455511	NA
6	5	1.CaboSanLucas	GUACHINANGO	16	2005	673552	NA
7	6	1.CaboSanLucas	GUACHINANGO	16	2006	672552	NA
8	7	1.CaboSanLucas	GUACHINANGO	16	2007	487502	NA
9	8	1.CaboSanLucas	GUACHINANGO	16	2008	412228	NA
10	9	1.CaboSanLucas	GUACHINANGO	16	2009	411658	NA
11	10	1.CaboSanLucas	GUACHINANGO	16	2010	403512	NA

- **File 2** is a metadata file with information about the stocks included in timeseries **File 1**. It has many columns (not all which are pictured here), from stock name to fishery start/end date, to estimated starting biomass, etc. It also includes a bunch of parameters that we're going to try to estimate, such as F_{lim}, B_{lim}, B_{msy}, etc. But before we can estimate any of these using **"SI_4_MSY_gulf_of_california.R"**, there are some columns that we need to fill out first.

  NOTE: Delete the stocks that don't have enough data from **File 2** (it doesn't matter if you keep them in **File 1**). In general, delete stocks which have less than 10 years of data.



	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	
1	roup	Name	EnglishNam	ScientificNar	Source	MinOfYear	MaxOfYear	StartYear	EndYear	Flim	Fpa	Blim	Bpa	Bmsy	FMSY	M
2	UACHINAN	GUACHINAN	GUACHINAN	GUACHINAN	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
3	ANGOSTA	LANGOSTA	LANGOSTA	LANGOSTA	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
4	IBURON	TIBURON	TIBURON	TIBURON	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
5	ARGO	PARGO	PARGO	PARGO	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
6	ABRILLA	CABRILLA	CABRILLA	CABRILLA	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
7	IERO	MERO	MERO	MERO	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
8	UREL	JUREL	JUREL	JUREL	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
9	IERNA	PIERNA	PIERNA	PIERNA	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
10	LMEJA	ALMEJA	ALMEJA	ALMEJA	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
11	AZON	CAZON	CAZON	CAZON	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
12	ENGUADO	LENGUADO	LENGUADO	LENGUADO	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
13	ULPO	PULPO	PULPO	PULPO	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
14	ALAMAR	CALAMAR	CALAMAR	CALAMAR	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
15	AYA Y SIMI	RAYA Y SIMI	RAYA Y SIMI	RAYA Y SIMI	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
16	UACHINAN	GUACHINAN	GUACHINAN	GUACHINAN	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
17	ARGO	PARGO	PARGO	PARGO	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
18	ABRILLA	CABRILLA	CABRILLA	CABRILLA	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
19	UREL	JUREL	JUREL	JUREL	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
20	IBURON	TIBURON	TIBURON	TIBURON	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
21	IERNA	PIERNA	PIERNA	PIERNA	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
22	ANDERA	BANDERA	BANDERA	BANDERA	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
23	AZON	CAZON	CAZON	CAZON	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA
24	AYA Y SIMI	RAYA Y SIMI	RAYA Y SIMI	RAYA Y SIMI	CONAPESCA	2001	2016	2001	2016	NA	NA	NA	NA	NA	NA	NA

Other files used:

- **"initial_conditions.R"**
- **"biomass_priors.R"**
- **"iuu_correction.R"**
- **"select_int_year.R"**

Steps:

(1) Create **File 1**

(2) If necessary, update **File 1** using **"iuu_correction.R"**

(3) Create an empty **File 2**, and populate the columns as follows:

  (4) Populate columns A - M ("ID" through "EndYear") as you would expect.

  (5) Go to Fishbase or Sealife base and fill in the "resilience" estimates for each species/taxa listed in (column Z). **CHECK**

  (6) Use FAO data, reported in tonnes, to define starting biomass_low - "stb.low" (column AC) & starting biomass_high - "stb.high" (column AD). To accomplish this task, consult the priors generated by the **"biomass_priors.R"** code. This code draws on assumptions made by **Giron-Nava (2018) S1** and outputs biomass

priors for broad stocks such as "groupers" and "sharks" as opposed to, say, a specific species of grouper or shark. **CHECK**

(7) Use the interactive module generated by **"select_int_year.R"** to find a suitable estimate for the intermediate year - "int.year" (column AE). **CHECK??**

(8) Run **"initial_conditions.R"** code to estimate values for intermediate biomass - "intb.low" through end biomass (high) - "endb.high" (columns AF - AI).

(9) Don't worry about the additional columns (columns AJ - AO) at this time.

(10) Now, everything should be ready to go. Use **"SI_4_MSY_gulf_of_california.R"** to estimate parameters of interest, where:

L31. "catch_file" is **File 1**

L32. "id_file" is **File 2**

NOTE: Avoid heavy modifications to this file, but be sure to modify the path for your output file on L1326.

(11) See L228 of code - using this for loop, you can analyze all stocks or just a key few as dictated by their index numbers listed in "id_file".

(12) Pay attention to L268 and L270 as these lines assume that catches are reported in kilograms (kg).

(13) L301 transforms the "resilience" of each taxa into a numeric range.

(14) Pay attention to the lines of code which generate the output files listed at the top of the code (L35 & L36), i.e. L1079 & L1186.

(15) Using the final output file (generated on L1326), you can learn about B/Bmsy over time.

References:

Cisneros-Montemayor, A. M., Cisneros-Mata, M. A., Harper, S., and Pauly, D. (2013) Extent and implications of IUU catch in Mexico's marine fisheries. Marine Policy, **39**, 283-288. doi: 10.1016/j.marpol.2012.12.003

Giron-Nava, A., Johnson, A. F., Cisneros-Montemayor, A. M., and Aburto-Oropeza, O. (2018) Managing at Maximum Sustainable Yield does not ensure economic well-being for artisanal fisheries. *Fish and Fisheries,* **20(2)**, 214-223. doi: 10.1111/faf.12332

Froese, R., Demirel, N., Coro, G., Kleisner, K. M., and Winker, H. (2017) Estimating fisheries reference points from catch and resilience. *Fish and Fisheries*, **18(3)**, 506-526. doi: 10.1111/faf.12190.

See also: https://oceanrep.geomar.de/33076/